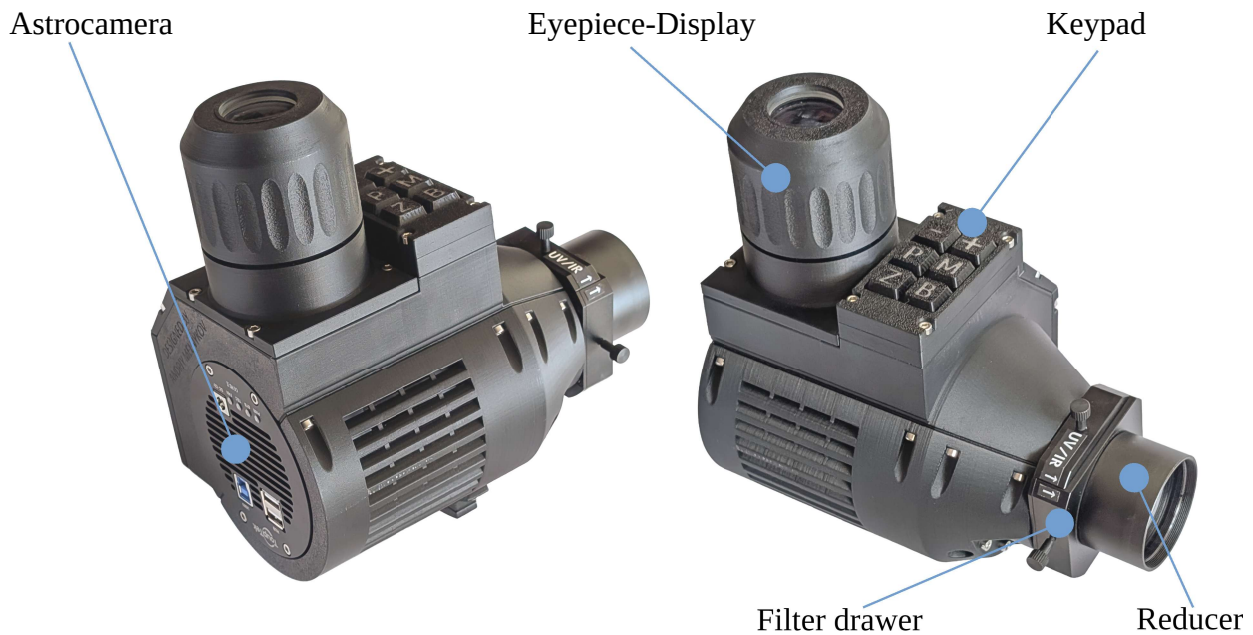


User manual for electronic eyepiece V3, 19.09.2026

The electronic eyepiece is designed for observing deep-sky objects (galaxies, nebulae, globular clusters and stars) with the CDK20 telescope.

The eyepiece is not suitable for observing the Moon or planets and **must never be used to observe the Sun!**

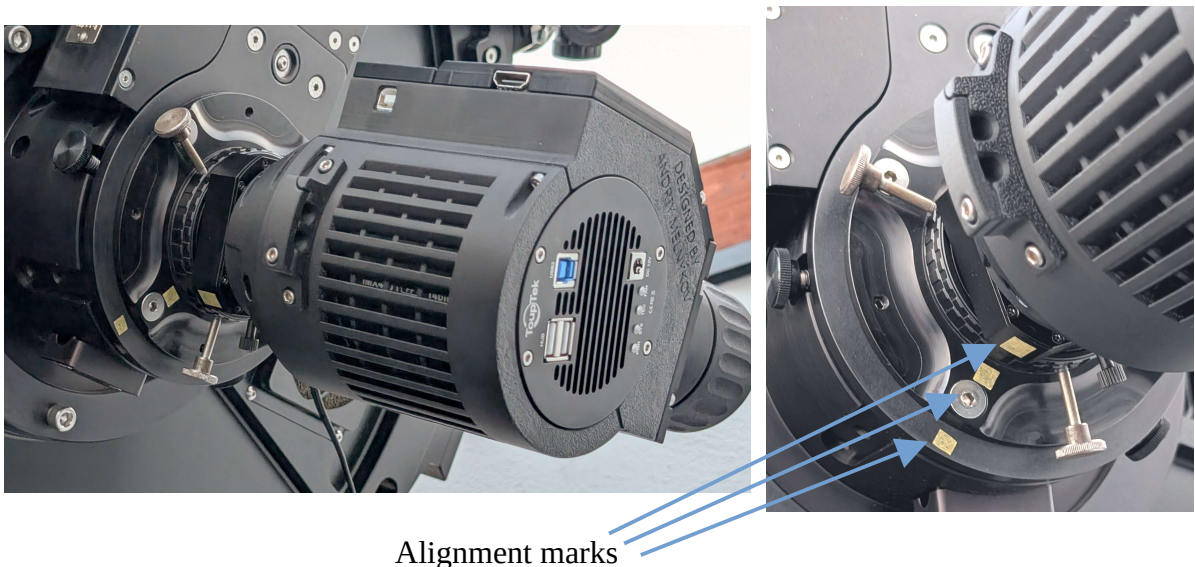
The electronic eyepiece consists of an astronomy camera, filter drawer, optical reducer (which also serves as a 2-inch barrel), eyepiece display and keypad. The camera, reducer and filter drawer form the permanently assembled optical unit. The outer housing with the display and keypad can rotate around the camera for more comfortable viewing.



Mounting

Insert the electronic eyepiece directly into the telescope's 2-inch holder (without a star diagonal) and secure it with the thumbscrews. Check the following:

- The bright alignment marks on the eyepiece and telescope must line up.
- The eyepiece must sit flush and must not be tilted.



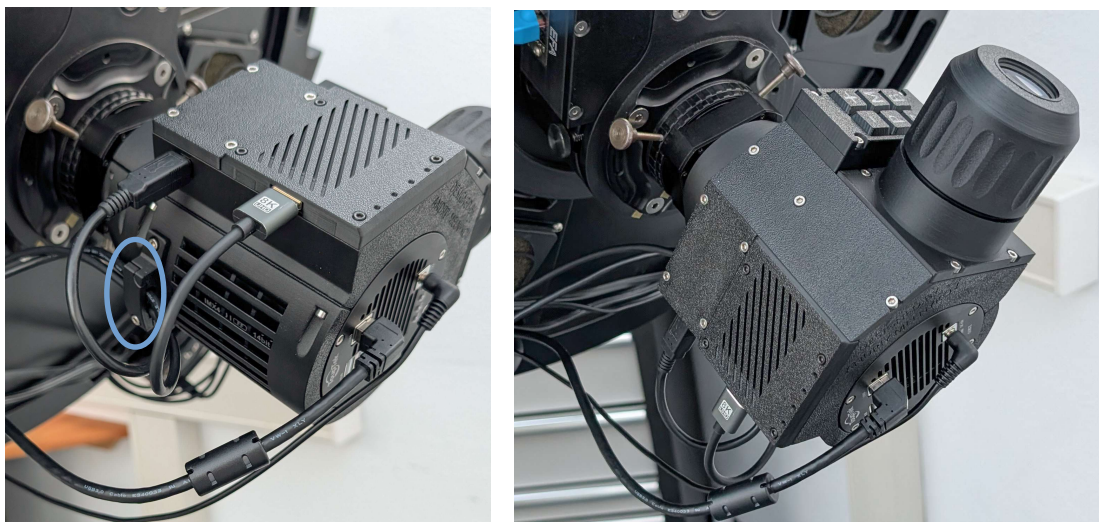
Connections

The electronic eyepiece requires four cables:

- Angled USB 3.0, camera → USB hub on the telescope:
- 12 V DC plug, camera → USB hub on the telescope:
- Straight USB 2.0, keypad → USB hub on the telescope:
- HDMI cable on the telescope → HDMI for the eyepiece display:



The camera and filter drawer must always remain attached to the telescope in the same position. The housing with the eyepiece display can be rotated around the camera with slight resistance. The USB keypad and HDMI cables can be secured to the housing to relieve strain on the connectors (these two cables move with the housing when the eyepiece is rotated):



Operation

The electronic eyepiece software is located on the computer in the “DigitalEyepieceAPP” folder on the desktop. The folder contains three shortcuts for launching the software with different settings.

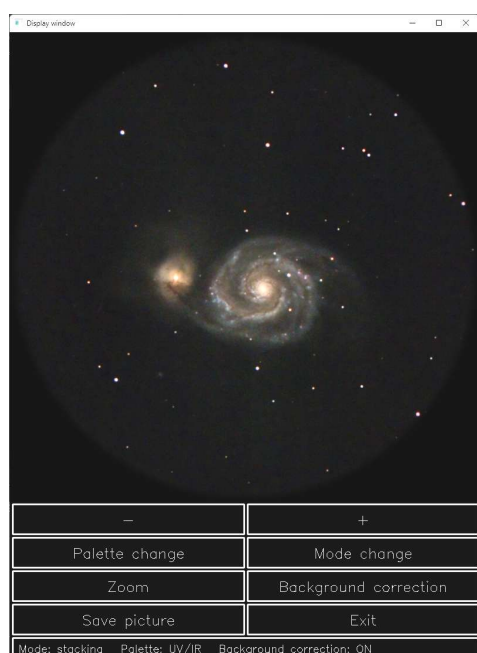
- “*DigitalEyepieceAPP*” starts the software in standard mode with a 0.5° field of view.

- “*DigitalEyepieceAPP_ZOOM*” starts the software in 2x zoom mode with a 0.25° field of view. This mode is less sensitive than standard mode.

- “*DigitalEyepieceAPP_HighSensitivity*” starts the software with a 0.5° field of view, higher sensitivity, lower resolution and less aggressive colors for very faint objects.

The software is controlled using a small number of GUI buttons on the PC or, alternatively, using the keypad on the eyepiece. The six buttons on the eyepiece correspond to the six upper GUI buttons in the software.

Images are saved automatically once per minute in the “SW/saved_pictures” folder.



-, +	-, +	Brightness, stretching, bring out details
Palette change	P	Switch color palettes for different filters
Mode change	M	Switch between real-time and stacking modes
Zoom	Z	Magnify the image
Background correction	B	Additional background correction
Save picture		Save image
Exit		Exit program

-, +: Adjust the image brightness and how much detail is shown in the deep-sky object. Start with a low setting to reduce visible noise, then increase it as stacking progresses to reveal more detail. There is always a trade-off between noise and detail.

Mode change, M: Switches between real-time and stacking modes. In real-time mode, you can move the telescope to centre the object. In stacking mode, the telescope must remain still and the mount must be tracking.

Zoom, Z: Enables or disables digital magnification, which can improve the view of small objects. It is also used for focusing in real-time mode. For higher-quality magnification, restart the software using the “_ZOOM” shortcut.

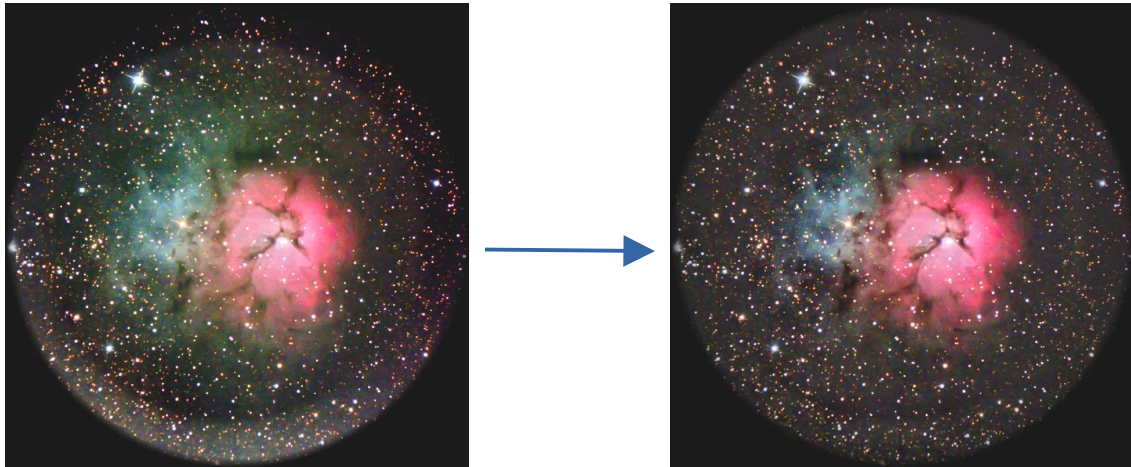
Palette change, P: Switches between three color palettes for different filters. The selected palette must match the filter in the filter drawer.

- „UV/IR“ – approximately reproduces natural colors.

- „DualBand“ – approximates the HHO palette, displaying H-alpha as orange and O-III as deep blue so that details are more visible.

- „UHC“ – reproduces natural colors as closely as possible with the UHC light-pollution filter.

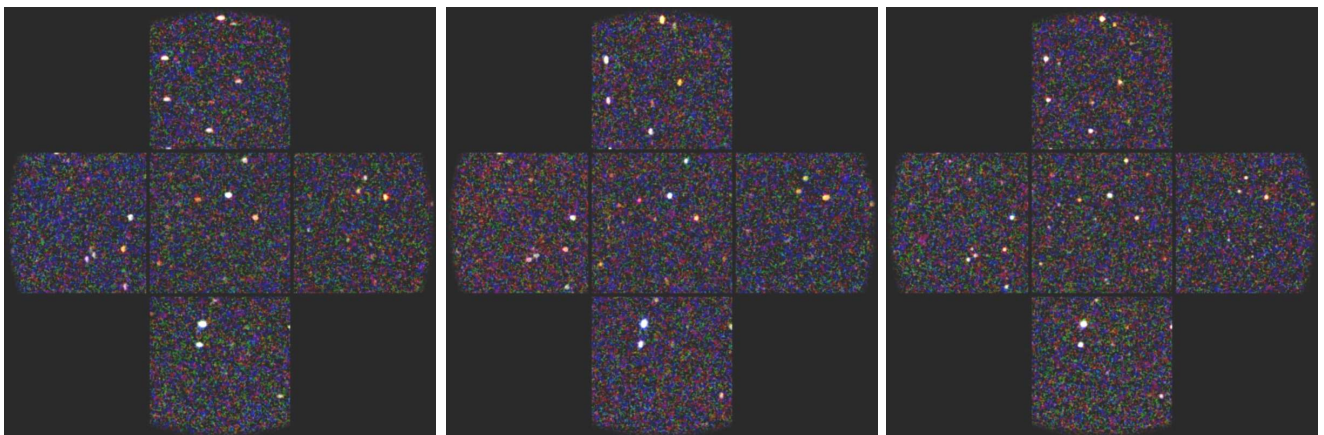
Background correction, B: Enables or disables additional background correction if the background appears blotchy, especially near the edge. This function works correctly only when the object is smaller than the field of view.



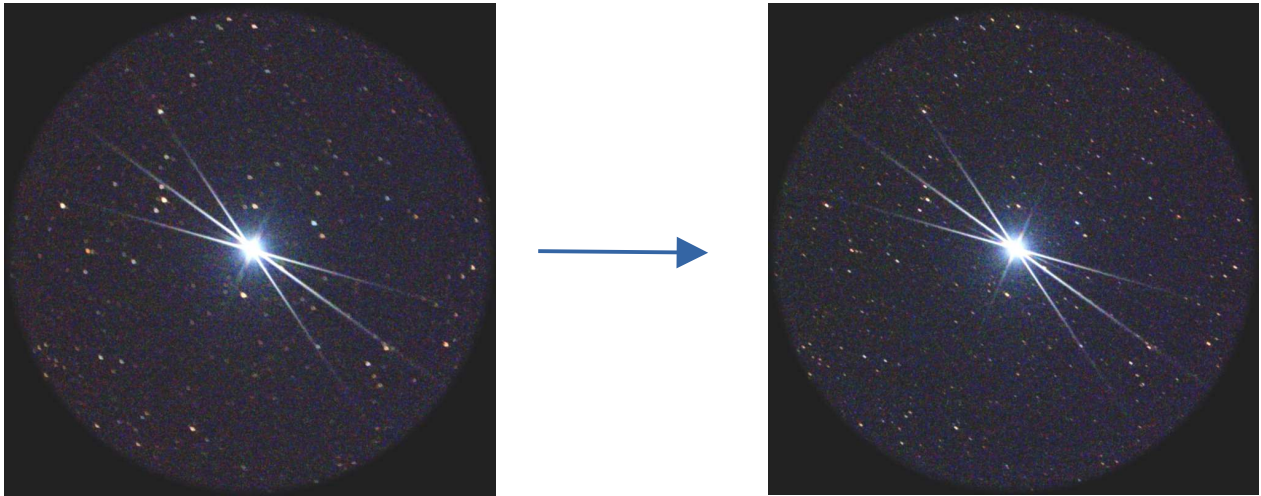
The status bar at the bottom of the window shows the active mode (real-time or stacking), the active color palette and whether background correction is enabled.

Procedure

1. Preferably with the mount in the “Park” position, attach the eyepiece to the telescope and connect all cables. When the computer is switched on, the eyepiece displays the background image as configured on PC.
2. Start the DigitalEyepieceAPP software. The PC and eyepiece should display a noisy or overexposed image, or a real-time image of stars if the telescope is pointing upwards.
3. Set the mount to the “Unpark” position and slew to a star field.
4. Focus in real-time mode with zoom enabled. The software displays enlarged sections from the centre of the image and from four areas near the edge. A feature of the reducer is that stars at the edge appear round only when the optimum focus has been reached. Use the shape and size of the stars as a focusing guide. Focus as usual with the remote control for the motorised focuser.



5. Alternatively, focus with a Bahtinov mask. Place the Bahtinov mask over the front of the telescope. Slew to a bright star and centre it in the eyepiece. Use the telescope's remote control to focus until the pattern appears symmetrical. The “Zoom” button can be used if desired.



Then remove the Bahtinov mask!

6. Focus the eyepiece display using the rotating focusing sleeve. It can be adjusted for each observer:



7. Select an object and slew the mount to it. If necessary, rotate the housing to place the eyepiece in a comfortable viewing position (do not loosen the thumbscrews). Watch the cables while rotating it. Centre the object and enable stacking mode.

8. To view a new object, enable real-time mode, slew to the object, rotate the eyepiece if necessary and then enable stacking mode.

9. A dual-band or UHC filter is useful when observing emission nebulae. When inserting a filter, align the arrow marks on the filter and eyepiece. The color palette selected in the software must match the installed filter. A retaining screw holds the filter in the filter drawer.